



Mark Scheme (Results)

Summer 2024

Pearson Edexcel International GCSE
In Mathematics B (4MB1) Paper 02

Edexcel and BTEC Qualifications

Edexcel and BTEC qualifications are awarded by Pearson, the UK's largest awarding body. We provide a wide range of qualifications including academic, vocational, occupational and specific programmes for employers. For further information visit our qualifications websites at www.edexcel.com or www.btec.co.uk. Alternatively, you can get in touch with us using the details on our contact us page at www.edexcel.com/contactus.

Pearson: helping people progress, everywhere

Pearson aspires to be the world's leading learning company. Our aim is to help everyone progress in their lives through education. We believe in every kind of learning, for all kinds of people, wherever they are in the world. We've been involved in education for over 150 years, and by working across 70 countries, in 100 languages, we have built an international reputation for our commitment to high standards and raising achievement through innovation in education. Find out more about how we can help you and your students at: www.pearson.com/uk

Summer 2024

Question Paper Log Number P76504A

Publications Code 4MB1_02_2406_MS

All the material in this publication is copyright

© Pearson Education Ltd 2024

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme.
Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.
- **Types of mark**
 - M marks: method marks
 - A marks: accuracy marks
 - B marks: unconditional accuracy marks (independent of M marks)

Abbreviations

- cao – correct answer only
- cso – correct solution only
- ft – follow through
- isw – ignore subsequent working
- SC – special case
- oe – or equivalent (and appropriate)
- dep – dependent
- indep – independent

- awrt – answer which rounds to
- eeoo – each error or omission
- cas – Correct answer scores full marks (unless from obvious incorrect working)
- wr – working required

No working

If no working is shown then correct answers normally score full marks

If no working is shown then incorrect (even though nearly correct) answers score no marks.

With working

If it is clear from the working that the “correct” answer has been obtained from incorrect working, award 0 marks.

If a candidate misreads a number from the question: eg. uses 252 instead of 255; follow through their working and deduct 2A marks from any gained provided the work has not been simplified. (Do not deduct any M marks gained.)

If there is a choice of methods shown, then award the lowest mark, unless the subsequent working makes clear the method that has been used

Examiners should send any instance of a suspected misread to review (but see above for simple misreads).

Ignoring subsequent work

It is appropriate to ignore subsequent work when the additional work does not change the answer in a way that is inappropriate for the question: eg. Incorrect cancelling of a fraction that would otherwise be correct.

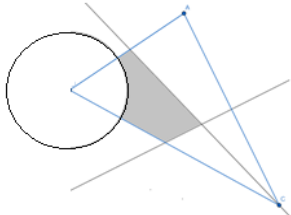
It is not appropriate to ignore subsequent work when the additional work essentially makes the answer incorrect eg algebra.

Transcription errors occur when candidates present a correct answer in working, and write it incorrectly on the answer line; mark the correct answer.

Parts of questions

Unless allowed by the mark scheme, the marks allocated to one part of the question CANNOT be awarded to another.

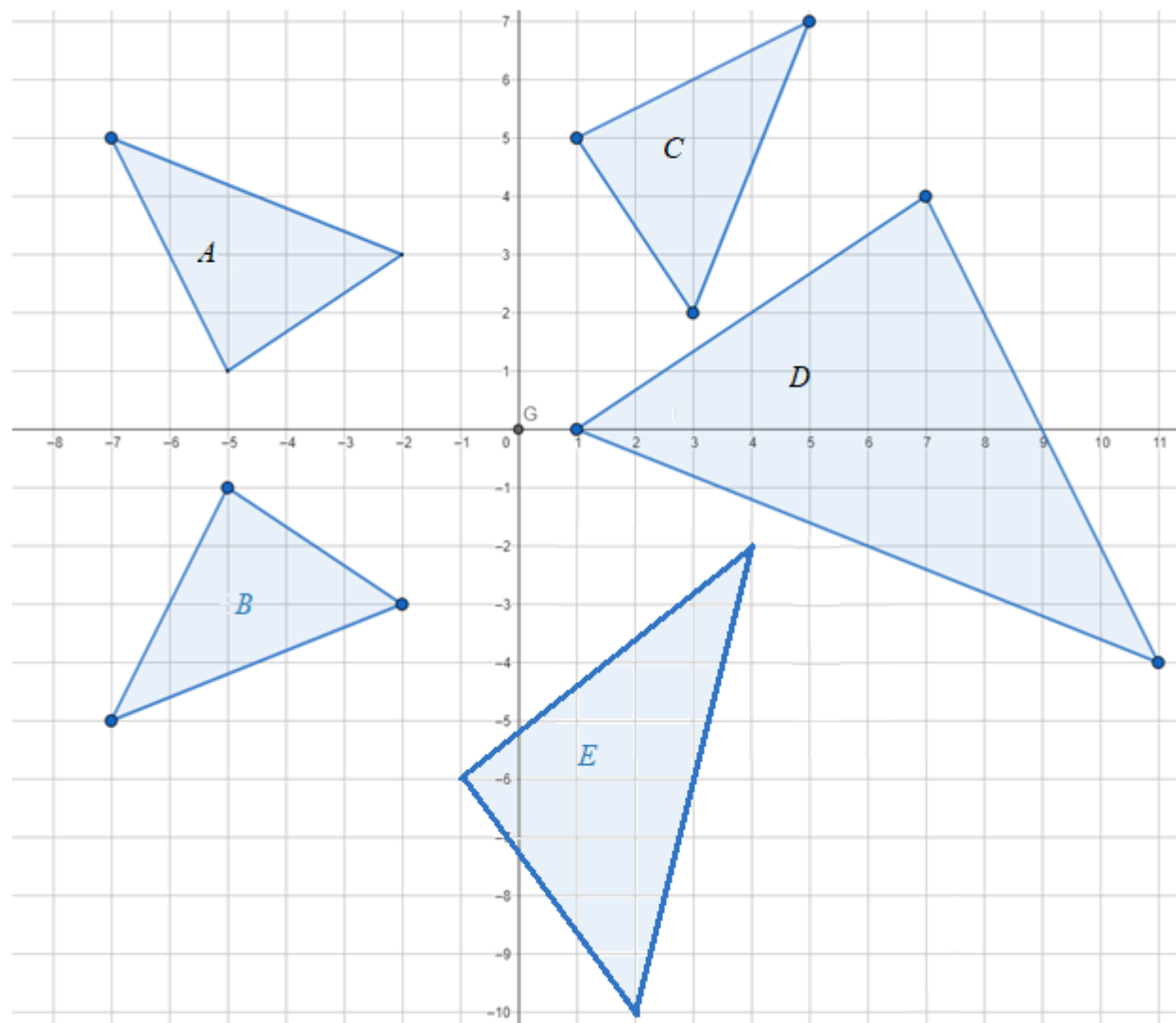
Question		Working	Answer	Mark	Notes
1	(a)		1.596×10^{-5}	1	B1 ISW
	(b)	$\frac{81460\,000\,000\,000}{3142.7} [= 2.5920... \times 10^{10}]$		4	M1 for use of Speed = distance / time. Condone missing zeros or incorrect standard form for numerator/denominator. Allow if clear trying to use 81 460 000 000 000 (eg 460 000 000 000) and 3142.7
		$\frac{"2.5920... \times 10^{10}"}{60 \times 60 \times 24} \left[= \frac{"2.5920... \times 10^{10}"}{86400} \right]$			M1 for using a correct method to change their Speed from km/day to km/second.
		300 004			A1 awrt 300000 Dependent on both M marks being awarded.
			3×10^5		A1 from correct working. Dependent on both M marks being awarded. Allow awrt 3.0×10^5
	ALT	$3142.7 \times 60 \times 60 \times 24 [= 2.715... \times 10^8]$			M1 for using a correct method to change 3142.7 days to seconds or allow for 271529280 given to minimum of 3sf
		$\frac{81460\,000\,000\,000}{"2.715... \times 10^8"}$			M1 for use of Time = distance / speed. Condone missing zeros or incorrect standard form for numerator/denominator. Allow if clear trying to use 81 460 000 000 000(eg 460 000 000 000) and their changed 3142.7
		300004			A1 awrt 300000 Dependent on both M marks being awarded.
			3×10^5		A1 from correct working. Dependent on both M marks being awarded Allow awrt 3.0×10^5
		wr			Total 5 marks

Question		Working	Answer	Mark	Notes
2	(a)		a circle centre B with radius 3 cm	1	B1 full circle drawn. Use overlay
	(b)		a correct <i>perpendicular</i> bisector of AC with arcs shown	2	B2 for a correct bisector with [2 pairs of arcs] or [2 arcs centre C and the cross drawn] (use overlay) must cross/touch AC NB bisector does not need to cross/touch BC (B1 for a bisector without arcs or only one pair drawn or correct arcs without bisector drawn. Arcs must cross within the lines on the overlay or would if they were extended
	(c)		a correct bisector with arcs shown	2	B2 for a correct bisector with arcs (use overlay) (B1 for a bisector without arcs or correct arcs without bisector drawn) NB bisector does not need to cross/touch AB
	(d)		correct region shaded	1	B1ft only follow through if they have a circle centre B of any radius (allow partial circle of any radius that crosses BA and BC) and at least B1 for bisector drawn in part (b) and part (c) with the 3 loci being a boundary of the area. The angle bisector must cross AB and the perpendicular bisector must cross BC . Only allow if the correct area is shaded.
					Total 6 marks

Question	Working	Answer	Mark	Notes
3	(a)			

Question	Working	Answer	Mark	Notes
In this question ignore any labelling on the triangles				
4	(a)		correct triangle <i>B</i> drawn	2 B2 correct triangle drawn (–2, –3) (–7, –5)(–5, –1) Award 2 marks for a correct triangle drawn irrespective of working in the working space. (B1 for 2 correct vertices plotted correctly or a triangle of the correct size and orientation, or 3 correct vertices listed)) SC B1 for a correct reflection applied to triangle <i>A</i> in the <i>y</i> -axis. [vertices (2, 3) (5, 1) (7, 5)] or for a correct reflection in the <i>x</i> – axis applied to triangle <i>D</i> [vertices (1, 0) (7, –4) (11, 4)]
	(b)		correct triangle <i>C</i> drawn	2 B2 correct triangle drawn (3, 2) (5, 7)(1, 5) (B1 for 2 points correct or for 90° clockwise rotation but incorrect position or for correct rotation of 90° around (0, 0) anticlockwise (–3, –2) (–1, –5)(–5, –7))
	(c)		Enlargement	3 B1 allow enlarge. Do not allow bigger B0 if multiple transformations stated. Multiple transformations are when more than one of reflection (mirrored), rotation (turn), translation (move), is stated eg a vector or SF or equation of a line do not imply multiple transformations
			Centre (–1, 2)	B1 for (–1, 2) Do not accept if written as a column vector.
			SF –2	B1 for –2
	(d)	$\begin{pmatrix} -1 & -1 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} -2 & -5 & -7 \\ 3 & 1 & 5 \end{pmatrix}$		3 M1 for the intention to multiply the right way. Points can be in any order. May be implied by writing the matrices in the correct order or correctly stating or plotting one point.
		$= \begin{pmatrix} -1 & 4 & 2 \\ -6 & -2 & -10 \end{pmatrix}$		M1 for at least two correct columns or correctly stating or plotting two points.
			correct triangle <i>E</i> drawn	A1 correct triangle drawn NB Award 3 marks for a correct triangle drawn irrespective of working in the working space.

	(e)	$\mathbf{M}^{-1} = \begin{pmatrix} -1 & 0.5 \\ 0 & -0.5 \end{pmatrix}$ or matrix for reflection in x-axis is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ or $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} -1 & 4 & 2 \\ -6 & -2 & -10 \end{pmatrix} = \begin{pmatrix} -2 & -5 & -7 \\ -3 & -1 & -5 \end{pmatrix}$		3	M1 for a correct inverse matrix for M or correct matrix for reflection in the x-axis or for setting up a correct matrix equation. Points in any order but the order must be the same in both matrices
		$\mathbf{N} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0.5 \\ 0 & -0.5 \end{pmatrix}$ or 2 of $\begin{bmatrix} -a - 6b = -2 \\ 4a - 2b = -5 \\ 2a - 10b = -7 \end{bmatrix}$ and 2 of $\begin{bmatrix} -c - 6d = -3 \\ 4c - 2d = -1 \\ 2c - 10d = -5 \end{bmatrix}$			M1 dep on M1 fully correct product or 2 correct equations from each set (any letters) May be implied by a correct answer
			$\begin{pmatrix} -1 & 0.5 \\ 0 & 0.5 \end{pmatrix}$		A1 cao
		cas			Total 13 marks



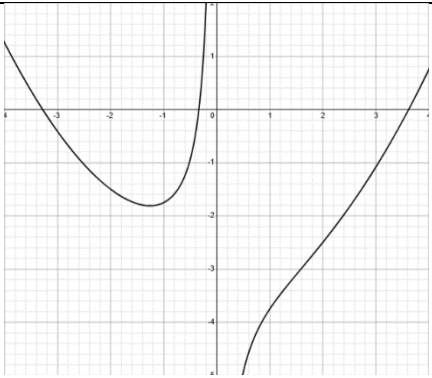
Question	Working	Answer	Mark	Notes
5	$\frac{dx}{dt} = 2kt - 3t^2$		5	M1 Attempt to differentiate ie $t^n \rightarrow t^{n-1}$ for at least one term
	$4k - 3 \times 4 = 24 [\Rightarrow k = 9]$ oe			M1 dep on the first M1 being awarded. subst $t = 2$ into their differential and equating to 24 leading to a value for $k \neq 0$
	$2 \times 9t - 3t^2 = 0 [\Rightarrow t = 0 \text{ or } t = 6]$			M1 dep on the first M1 being awarded. subst their value of k into their differential and equating to 0 leading to a value for $t > 0$
	$[x =] 9 \times 6^2 - 6^3$			M1 dep on 2 of the previous 3 method marks being awarded. subst their positive value of $t \neq 2$ and $k \neq 0$ into the equation for x . If t or k is incorrect working must be shown.
		108		A1 Do Not ISW
	cas			Total 5 marks

Question		Working	Answer	Mark	Notes
6	(a)		0.4	1	B1 allow $\frac{2}{5}$ oe
	(b)	"0.4"×"0.4"		2	M1 ft their answer to part (a) If no working shown you may need to check their answer follows form (a) for this mark eg if (a) is 0.6 award M1A0 for 0.36
			0.16		A1 allow $\frac{4}{25}$ oe
	(c)		84	1	B1
	(d)	$\frac{\frac{1}{2}n}{n} \times \frac{\frac{1}{2}n-1}{n-1} \times \frac{\frac{1}{4}n}{n-2}$ oe		6	M2 A fully correct expression. (M1 for setting up an expression with at least one of the numerators correct and all the denominators correct eg $\frac{\frac{1}{2}n}{n} \times \frac{\frac{1}{2}n-1}{n-1} \times \frac{\frac{1}{4}n}{n-2} \times m$ where m is an integer for M1)
		$\frac{1}{2} \times \frac{1}{2(n-1)} \times \frac{n}{4} = \frac{5}{78}$ oe or $39 \times (\frac{1}{2}n-1) \times \frac{1}{4}n = 5 \times (n-1)(n-2)$ oe			M1dep on M2 for forming a correct equation and realising that $(n-2)$ can be cancelled eg $\frac{n}{16(n-1)} = \frac{5}{78}$ or for forming a correct quadratic equation by removing all terms in n correctly from the denominator eg $\frac{39}{4}n^2 - \frac{39}{2}n = 10n^2 - 30n + 20$ May be implied by $n = 40$
		$78n = 80n - 80$ or $n^2 - 42n + 80 = 0$			M1 dep on M3 for removing the fractions to get a correct linear equation, or a correct 3 term quadratic equation or $n = 40$
		$\frac{\frac{1}{4} \times "40"}{"40"} \times \frac{\frac{1}{4} \times "40" - 1}{"40" - 1} \times \frac{\frac{1}{4} \times "40" - 2}{"40" - 2}$			M1 A correct method to the required probability. Only ft their value of n if a positive integer and working shown. May be implied by $\frac{3}{247}$ or 0.05 th m1 which 12[145...]
		cas	$\frac{3}{247}$		A1 or equivalent fraction. Do Not allow written as a decimal
		PTO for ALT for (d)			

ALT	(d)	$\frac{2x}{4x} \times \frac{2x-1}{4x-1} \times \frac{x}{4x-2} \text{ oe or}$ $\frac{y}{2y} \times \frac{y-1}{2y-1} \times \frac{\frac{1}{2}y}{2y-2} \text{ oe}$			M2 A fully correct expression. x is the number of blue/green counters, y is the number of red counters (M1 for setting up an expression with at least one of the numerators correct and all the denominators correct eg $\frac{2x}{4x} \times \frac{2x-1}{4x-1} \times \frac{x}{4x-2} \times m$ where m is an integer for M1)
		$\frac{1}{2} \times \frac{1}{4x-1} \times \frac{x}{2} = \frac{5}{78} \text{ or}$ $\frac{1}{2} \times \frac{1}{2y-1} \times \frac{y}{4} = \frac{5}{78} \text{ or}$ $39 \times (2x-1) \times x = 5 \times (4x-1)(4x-2) \text{ or}$ $39 \times y = 20 \times (2y-2)$			M1dep on M2 for forming an equation and realising that $(2x-1)$ or $(4x-2)$ or $(y-1)$ can be cancelled or for forming a quadratic equation by removing all terms in x or y correctly from the denominator May be implied by $x = 10$ or $y = 20$
		$78x = 80x - 20 \text{ or } 78y = 80y - 40 \text{ or}$ $2y^2 - 42y + 40 = 0 \text{ or } 2x^2 - 21x + 10 = 0$			M1 dep on M3 for removing the fractions to get a correct linear equation, or a correct 3 term quadratic equation or $x = 10$ or $y = 20$
		$\frac{"10"}{4 \times "10"} \times \frac{"10"-1}{4 \times "10"-1} \times \frac{"10"-2}{4 \times "10"-2} \text{ or}$ $\frac{\frac{1}{2} \times "20"}{2 \times "20"} \times \frac{\frac{1}{2} \times "20"-1}{2 \times "20"-1} \times \frac{\frac{1}{2} \times "20"-2}{2 \times "20"-2}$			M1 a correct method to the required probability. Only ft their value of x or y if a positive integer and working shown. May be implied by $\frac{3}{247}$ or 0.012[145...]
			$\frac{3}{247}$		A1 or equivalent fraction. Do Not allow written as a decimal
		cas			Total 10 marks

Question	Working	Answer	Mark	Notes
7	(a)		3	M1 for a fully correct method or expression to find BF eg $\frac{BC}{\sin 60} = \frac{40}{\sin 30}$ and $BF^2 = 40^2 + "4800"$ (NB $BC^2 = 4800$ and $BC = 40\sqrt{3}$ or 69.2[82...]) Allow any letters for BF
				M1 dependent on M1 being awarded. for a fully correct method to find EH^2 or EH eg $[EH^2 =]100^2 + (22.5)^2$ or $[EH =]\sqrt{100^2 + (22.5)^2}$ condone $100^2 + (57.5 - "80")^2$
				A1 oe dep on both previous M marks being awarded. for a fully correct expression for EH^2 or EH seen leading to 102.5 eg $[EH^2 =]100^2 + (22.5)^2$ or 10506.25 or $[EH =]\sqrt{100^2 + (22.5)^2}$ or $\sqrt{10506.25}$ or $\frac{100}{\cos(12.68...)}$
	(b)			X is a point on the line FC such that HX is horizontal and Y is a point on the line BC such that HY is vertical
			5	M1 for a fully correct method to find HX (the distance from H to FC) eg $("80" - 57.5)\sin 60$ Allow 19.4 or 19.5 Allow awrt 19 from correct working
				M1 for a fully correct method to find CX (the height of H above BC) Allow 29 from correct working
				M1 dep on both previous M marks being awarded. For a fully correct method to find DH^2 or DH ft their HX and CX if clearly labelled or comes from correct working. Allow 11000 from correct working or awrt 105 or 106
				M1 dependent on all 3 method previous M marks being awarded. For a fully correct method to find $\cos \angle EHD$. For 11206.25 allow (awrt 105 or awrt 106) ²
		cas	22.1	A1 awrt 22.1 Allow 22.2 or 22 or 22.0 from correct working
				Total 8 marks

Question	Working	Ans	Mark	Notes
8	(a) $\frac{54}{6} \times 4$ oe		2	M1 for a correct method to find number of watches May be seen within a ratio ie 36 : 54 : 45 or 36 : 54 or 36 : 45 Condone $\frac{54}{15} \times 4$ May be implied 14.4 for M1
		36		A1
	(b) $120 \times 32 + 0.06 \times 13000 [= 3840 + 780]$		2	M1 for a complete method to find Mel's earnings.
		4620		A1
				SC M1A0 for 5580 (misread 150)
	(c) $8040 - 150 \times 32 [= 3240]$ or $0.6w + 150 \times 32 = 8040$		3	M1 a correct expression to find the amount earnt on watches or a correct equation
	$\frac{"3240"}{0.06}$ or $0.06w = "3240"$			M1 dep on M1 being awarded Allow 6% = "3240" or 0.06% = "3240"
		54000		A1
	(d) $0.64 \times 250 \times 140 [= 160 \times 140 = 22400]$		5	M1 for calculating the total selling price for watches sold in Swiss francs
	$[\$216 =] \frac{216}{1.08} [= 200]$			M1 for a correct method to change \$ to Swiss francs. May be seen in any calculation (need not be correct calculation) eg $(250 - 0.64 \times 250) \times \frac{216}{1.08}$ or $(250 - 160) \times \frac{216}{1.08}$ Implied by 18000
	$(250 - 0.64 \times 250) \times "200" [= 90 \times "200" = 18000]$			M1 for calculating the total selling price for watches sold in America in Swiss francs.
	"22400" + "18000" - 250×80 Or $40400 - 250 \times 80$			M1 If working shown allow "their $0.64 \times 250 \times 140" + n - 250 \times 80$ or $m + "their (250 - 0.64 \times 250) \times \frac{216}{1.08}" - 250 \times 80$ where m and n are any number. This may be embedded in a calculation eg $\frac{40400 - 20000}{20000} [\times 100 = 102]$ or implied by a profit of 102[%]
		20400		A1 do not ISW
	cas			Total 12 marks

Question	Working	Answer	Mark	Notes
9	(a)	$-1.5, -1.75, -3.75$	2	B2 All 3 values correct (B1 2 values correct or if have $-1.5, -1.8, -3.8$)
	(b)		3	B1 correct shape between $x = -0.5$ and $x = 0$ Points not joined and the curve must go above the x – axis between -0.5 and 0 and not cross the y -axis. B2 At least 8 points plotted correctly and joined with a reasonably smooth curve to form the 2 parts. Condone the points at $x = -0.5$ and $x = 0.5$ being joined to form 1 curve (ie condone crossing the y -axis) Do not allow straight lines for B2 (B1 for at least 6 points plotted correctly) NB for the 2 points at $x = 0.5$ and 3 check they are close to -1
	(c)	$3.6, -3.3, -0.3$	1	B1 Allow ± 0.1 Do not allow extra values
	(d)	Line $y = -1$ drawn	4	M1 for the line $y = -1$ drawn on the grid from -4 to 4
		$-2.5, -0.5, 3.1$		A1 dep on M1 For 3 correct values allow ± 0.2 must agree with their horizontal line and graph. Condone incorrect extras. May be seen as part of inequalities or coordinates A0 if values are given to 3dp or more For reference $x = -2.534070197, -0.517304045, 3.051374242$
		$-2.5 \leq x \leq -0.5$ or $0 < x < 3.1$		M1 dependent on a horizontal line being drawn at $y = -1$ One correct range using the correct values ± 0.2 which must agree with their horizontal line and graph. Allow $<$ for $\leq$ and $\leq$ for $<$ SC If drawn $y = 1$ allow for a correct inequality $-3.9 \leq x \leq -0.3$ or $0 < x < 4.1$ (all values ± 0.2)
		$-2.5 \leq x \leq -0.5$ and $0 < x < 3.1$		A1 dep on first M1 Allow $x \geq -2.5$ [and] $x \leq -0.5$ for $-2.5 \leq x \leq -0.5$ Allow $x > 0$ [and] $x < 3.1$ for $0 < x < 3.1$ must have $0 <$ but the others we will allow $<$ or $\leq$ or $>$ Allow other notation eg $[-2.5, -0.5]$ or $(0, 3.1]$ allow curved brackets for square brackets NB Allow $-2.5, -0.5$ and 3.1 all ± 0.2 but must have 0
				Total 10 marks

Question	Working	Answer	M	Notes
10 (a)	$6 \times \left(-\frac{1}{2}\right)^3 - (6p-1)\left(-\frac{1}{2}\right)^2 - (5p+q)\left(-\frac{1}{2}\right) + 2$		2	M1 for substitution of $x = -0.5$ into equation. Allow one slip of 0.5 indicated rather than -0.5
	wr $\text{eg } -\frac{3}{4} - \frac{3}{2}p + \frac{1}{4} + \frac{5}{2}p + \frac{1}{2}q + 2 = 0 \therefore 2p + q = -3$ $\text{or } -3 - 6p + 1 + 10p + 2q + 8 = 0 \therefore 2p + q = -3$			A1 dep on M1 awarded. For equating to zero (may be implied). A fully correct solution with at least one line of working with all the brackets removed. Do not allow $2p + q - 3 = 0$
(b)	$6p^3 - (6p-1)p^2 - (5p+q)p + 2$ or $6x^3 - (6p-1)x^2 - (5p+(-3-2p))x + 2$ or $6x^3 - \left(6\left(\frac{-3-q}{2}\right) - 1\right)x^2 - \left(5\left(\frac{-3-q}{2}\right) + q\right)x + 2$ or		7	M1 Substituting p in for x or $-3-2p$ for q or $\frac{-3-q}{2}$ for p or for correct remainder $-4p^2 - pq + 2$ from long division. This needs to be a correct expression. We will condone the use of an incorrect equation found in part(a)
	$6p^3 - (6p-1)p^2 - (5p+(-3-2p))p + 2 [=0]$ or $6\left(\frac{-3-q}{2}\right)^3 - \left(6\left(\frac{-3-q}{2}\right) - 1\right)\left(\frac{-3-q}{2}\right)^2 - \left(5\left(\frac{-3-q}{2}\right) + q\right)\left(\frac{-3-q}{2}\right) + 2 [=0]$ or $-4p^2 - p(-3-2p) + 2$ or $-4\left(\frac{-3-q}{2}\right)^2 - \left(\frac{-3-q}{2}\right)q + 2 [=0]$			M1 Substituting p in for x and $-3-2p$ for q or $\frac{-3-q}{2}$ for p or subst $-3-2p$ for q or $\frac{-3-q}{2}$ for p into their remainder to form an expression in terms of either p or q . Allow substitution into their expression (allow if subst into an incorrect expression) We will condone the use of an incorrect equation found in part(a)
	$2p^2 - 3p - 2 [=0]$ oe or $q^2 + 9q + 14 [=0]$ oe			A1 dependent on both previous M marks being awarded correct 3 term quadratic in terms of p or q
	$(2p+1)(p-2) [=0]$ or $(q+7)(q+2) [=0]$			M1 dep on at least 1 of the previous method marks being awarded for solving their 3 term quadratic equation by factorisation or correct use of formula/ completing the square. If the equation is incorrect working must be shown. (eg substitution into the formula with no simplification)
	$p = 2$ or $q = -7$			A1 dep on the previous A1 being awarded. For a correct value of p or q This implies 4th M1
	$2 \times "2" + q = -3$ oe $[q =] -3 - 2 \times "2"$ or $2p + (" - 7") = -3$ oe $[p =] \frac{-3 - (" - 7")}{2}$			M1 dependent on at least 1 method mark being awarded. For substituting their value of p into a correct equation to find q or their value of q into a correct equation to find p . Allow written as an expression. Working must be shown if the value of p or q is incorrect. Allow for a correct equation in terms of the second variable. Implied by $p = 2$ and $q = -7$ We will condone the use of an incorrect equation found in part(a)
	wr $p = 2$ and $q = -7$			A1 dep on the 1 st M1, 2 nd M1 and 1 st A1 being awarded. Correct values with $p = -0.5$, $q = -2$ both eliminated

	(c)	$(2x+1)(x-2)\left(3x-\frac{2}{2}\right)$		2	M1 ft their value of p must be of the form $(2x+1)(x-p)\left(3x-\frac{2}{p}\right)$
		<i>cas</i>	$(2x+1)(x-2)(3x-1)$		A1 cao do not ISW. Mark final answer. NB $\left(x+\frac{1}{2}\right)(x-2)\left(x-\frac{1}{3}\right)$ is M0A0
					Total 11 marks

Question		Working	Answer	Mark	Notes
11	(a)	$2m - 5 = 0$		2	M1 Setting $f(m) = 0$ allow any letter for m
		cas	2.5		A1 oe eg $5/2$ or $f(2.5)[=0]$
	(b)	cas	-15	1	B1oe cao
	(c)	cas	$-\frac{1}{2}$	1	B1oe allow $x = -0.5$ or $x \neq -0.5$ ISW DO NOT allow $x < -0.5$ or $x > -0.5$ Must be a single value
	(d)	$(2x - 5)(2x + 1) = 7 - 8x$ oe		4	M1 for a correct equation with the fraction removed. Brackets may be expanded. Condone missing brackets if recovered. If only seen expanded allow 2 sign errors.
		$4x^2 + 2x - 10x - 5 = 7 - 8x$			M1 dependent on M1 awarded. multiplying out brackets. Allow one incorrect term.
		$x^2 = 3$			M1 gaining a value for x^2 ft their equation if the x terms cancel. May be implied by correct answer
		wr	1.73		A1 dependent on the 1 st and 2 nd M1 being awarded. awrt 1.73 or $\sqrt{3}$ (condone $\pm$)
	(e)	$2(x^2 + 2x) - 5$ or $2\left(x^2 + 2x - \frac{5}{2}\right)$		3	M1 allow for $a = 2$ or $b = 1$ or $c = 7$ Allow these seen in an expression of the form $a(x + b)^2 - c$ This may be implied by the 2 nd M1
		$\left[2(x + 1)^2 - 2\right] - 5$ or $2\left[(x + 1)^2 - 1 - \frac{5}{2}\right]$			M1 allow for 2 of $a = 2$ or $b = 1$ or $c = 7$ Allow these seen in an expression of the form $a(x + b)^2 - c$
		cas	$2(x + 1)^2 - 7$		A1 if a correct expression is not seen we allow $a = 2$ and $b = 1$ and $c = 7$ clearly stated
	(f)	$\frac{y + "7"}{"2"} = (x + "1")^2$ or $\frac{x + "7"}{"2"} = (y + "1")^2$ oe		2	M1 Allow any 2 different letters to be used ft their $a \neq 1$, b and c (a , b and $c \neq 0$) from part (e) for $\frac{y \pm "c"}{"a"} = (x + "b")^2$ oe We will allow $\pm$ "their c "
		cas	$\sqrt{\frac{x + "7"}{"2"}} - "1"$		A1ft their a , b and c from part (e) Must be in terms of x (if $\pm$ given award M1 only eg $- "1" \pm \sqrt{\frac{x + "7"}{"2"}}$)
					Total 13 marks

